LESSON 4.1: Troubleshooter

What this lesson is about

This lesson introduces you to the Sterling B2B Integrator system troubleshooter utility. The troubleshooter is a graphical interface to system performance indicators. This lesson covers the components that are provided by the troubleshooter and the operator panel.

What you should be able to do

After completing this lesson, you should be able to:

- View database usage, services, and pools.
- View business process Queue, Usage statistics, and State information.
- Resume, restart, or terminate a business process.
- View system classpath and JNDI tree information.
- View the cleanup process.
- View environment statistics, including cache and memory used.
- View Remote Perimeter Server information if present.
- View Troubleshooter information about Sterling B2B Integrator cluster nodes.
- Monitor node information and threads with Cluster Monitor.

How you will check your progress

- The progress of the lesson is analyzed based on the successful application of the topics in the scenario and test the exercise independently.

References

Documentation

- Monitoring Operations:

  https://www.ibm.com/support/knowledgecenter/en/SS3JSW_5.2.0/com.ibm.help.performance_mgmt.doc/SIPM_MO_Overview.html

Troubleshooter Overview

Overview

The Troubleshooter tool is a graphical utility for monitoring system operation and diagnosing Sterling B2B Integrator system problems. Troubleshooter allows to view the system information about different nodes in a clustered environment, including DB usage and services, business process queue and usage information, and environment statistics.

Use the System Troubleshooter page to review system information and to start troubleshooting system issues in Sterling B2B Integrator.

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Troubleshooter Overview

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Select Node

The select node list allows to select a node in a clustered environment, triggering which node information to display in the System troubleshooting page.

Example

If you have two nodes in a cluster, Node 1 and Node 2 and you want to view the System troubleshooting page for Node 2, select Node 2 from the list and the System troubleshooting page for Node 2 displays. If you want to view Node 1 information, select Node 1 from the list and the system statistics and troubleshooting information for Node 1 displays.

Note

The Select Node list displays only if you are working in a clustered environment. Your selection mandates which node information displays in the remainder of the System troubleshooting page.

System Troubleshooting - System Status

System Status Overview

The System Status area shows the following information:

Option	Description
Stop the System	Stops the whole Sterling B2B Integrator cluster by using the script. After that, you still need to call the hardstop script to stop the whole system.
Select Node	The Select Node list is displayed only if you are working in a clustered environment. Displays the selected node information in the rest of the System troubleshooting page.
Host Information	The Host information displays the following information: Start time Uptime Host Location State Memory available Active threads
Classpath	Displays the Sterling B2B Integrator classpath.
JNDI Tree	Displays the JNDI tree in Sterling B2B Integrator.
Soft Stop	Stops a node of Sterling B2B Integrator by using the script interactively through the UI.
Database Usage	Displays the database space usage, database services (business process eligibility for archive, index, and purge), and environment pool usage.

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System Troubleshooting - System Status

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System Status Overview

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Option	Description
Business Process Queue Usage	Displays business process queue usage statistics such as cache disk usage, cache memory usage, queue statistics, and cache statistics.
Business Process Usage	Displays count of business processes by its state.
Cache Usage	Displays size and hit rate for object caches.
Threads	Displays active processes at a thread level.
Clean-Up Processes Monitor	Displays the time since the archive, index, purge, and recovery tasks were completed.
Controllers	The state and name of each controller or server in the Sterling B2B Integrator installation.
Adapters	Displays list of all the adapters in the system and their status.
Perimeter Server Status	The Perimeter Servers area displays the following information: Cluster node name (in a clustered environment only) Whether the perimeter server is on or off State, either enable or disable Name of the perimeter server Last activity

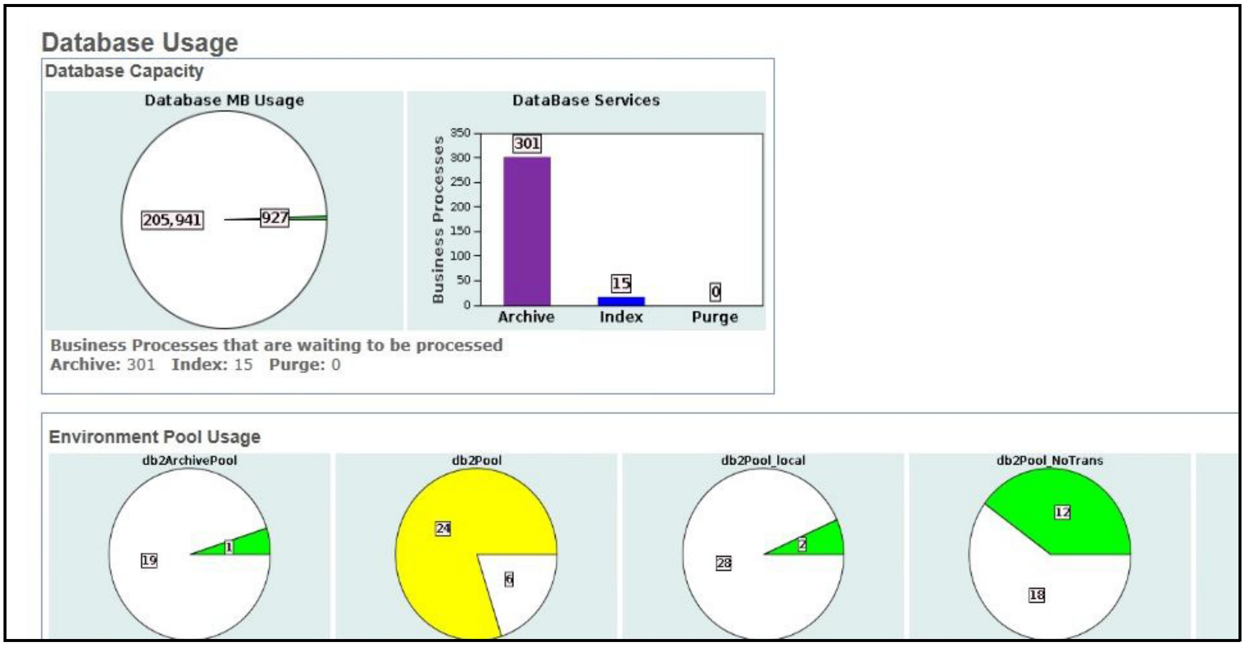
Note

System status information is collected and reported in real time. To view the most current system status, click **Refresh Status.**

Database Usage

Viewing Database Usage

To view database usage statistics, click **Database Usage** in System Troubleshooting page. The top of the Database Usage page shows the Database Capacity statistics, as shown in the screen capture:



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Database Usage

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Viewing Database Usage

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The Database Usage page includes information about the:

- Database Capacity
  - Database MB Usage - Size of the DB and the amount of the DB space that is used in megabytes.
    - Green indicates normal range
    - yellow is a warning
    - red is critical

Note

If the DB is set to auto-extend the reading can be 99% full all of the time, as it grows incrementally until allocated disk space is exceeded.

  - Database Services - Number of business processes that are waiting to be archived, indexed, or purged.
  - Environmental Pool Usage

Sterling B2B Integrator uses pools to store database connections. The bottom portion of the Database Usage page shows Environmental Pool Usage information. The database portion of the pool names changes depending on the database you are using.

- Database Access Test (runs whenever DB Usage screen is accessed)
  - Average Insert Time - Time the Database Access test took to test the number of database inserts indicated. This value must be monitored over time.
  - Number Inserts - Number that is performed to the database in the test. You can change the value in the dbAccessLoopCnt property in the install_dir/properties/ui.properties.in file and then run the script.
  - Insert Size - Size of inserts to the database that were used in the test. You can change the value in the dbAccessDataSize property in the install_dir/properties/ui.properties.in file and then run the script.

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Database Usage

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Viewing Database Usage

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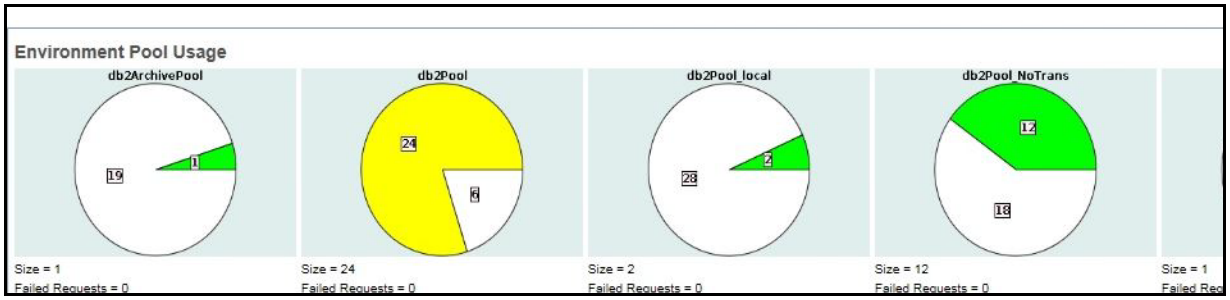
Example

The database is db2, you see db2Pool, as shown in the following figure.



Size is number of connections currently in the pool (not necessarily the max pool size). Failed Requests is the number of requests that must wait for a pool connection. If the pool is at max connections (as defined in tuning.properties) and Failed Requests is greater than zero, then pool size needs to be increased or Sterling B2B Integrator performance suffers.

To change pool settings:

1. Run the Performance Tuning Utility or manually edit the properties files.
2. Run the script.
3. Restart the system.

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Database Usage

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Viewing Database Usage

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The following pool settings are found in jdbc.properties. Pools DB2Pool and DB2Pool_NoTrans are dynamic variables that are linked to tuning.properties.

Pool	Description
DB2UIPool	User interface connections
DB2Pool_NoTrans	Non-transactional pool connections - MAX_NONTRANS_POOL as defined in tuning.properties
DB2Pool_local	jdbc.properties
DB2Pool	Transactional pool connections - MAX_TRANS_POOL as defined in tuning.properties
DB2ArchivePool	jdbc.properties

Exercise 4.1.1: Pool Connections

Introduction

Alice adds a database pool connection in the Sterling B2B Integrator environment.

This activity is good rwhen users are building a business process to access information that is stored in the database. The benefit to your environment is to reduce resource contention and decrease the amount of database failed request. Also, the system is not competing for resources that are used internally by the Sterling B2B Integrator software.

This exercise includes the following tasks:

- Using the Resource Manager to import a code list.
- Review the imported code list.
- Adding a pool connection to the system by editing the jdbc_customer.properties.in and running .
- Creating a Lightweight JDBC adapter service configuration with the new pool connection.
- Creating a business process includes a Lightweight JDBC adapter and an Assign service.

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Exercise 4.1.1: Pool Connections

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Managing Resources

In Sterling B2B Integrator, resources are files, templates, and documents that are deployed in Sterling B2B Integrator to test various actions. Sterling B2B Integrator enables to import and export resources, which can save time and increase the accuracy of duplicating resources on various Sterling B2B Integrator systems that are set up for unique purposes. Specifically, the Import and export options enable to import resources from:

- A Sterling B2B Integrator test environment to a Sterling B2B Integrator production environment
- One Sterling B2B Integrator system to another

Note

To import and export resources from one Sterling B2B Integrator environment to another, both the environments can be in same version. The resources can be imported from lower version environment to higher version environment but from higher version environment to lower version environment is not compatible.

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Exercise 4.1.1: Pool Connections

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Managing Resources

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You can import and export the following resource types:

■ Accounts	■ Application Configurations (only SAP)
■ Business Processes	■ Communities
■ ebXML Specifications	■ Maps
■ Mailboxes	■ PGP Profiles
■ Perimeter Servers	■ Report Configurations
■ Schedules	■ XML Schemas
■ Service Configurations	■ Trading Partner Data
■ Web Resources	■ WSDL
■ Web Templates	■ XSLTs

Instructions

Importing Information into the Database

Complete the following steps to import data into database:

Step 1: From the web browser, access **http://192.168.40.100:9000/dashboard** URL by specifying **admin** for User ID and **password** for Password.

Step 2: Click **Sign In**.

Step 3: From the **Administration Menu**, click **Deployment** > **Resource Manager** > **Import/Export**.

Step 4: In the Import/Export Resources page, under Import, next to the Import Resources, click **Go!**.

Step 5: In the **Import Resources: Import File** page, browse to the **Lab Files** and select **swift_code_list.xml**. Leave the blank and click **Next**.

Step 6: Click **Next**, do not create resource tag.

Step 7: Click **Yes** to update object in database, click **Next**.

Step 8: Move **all six** items to the bottom window and click **Next**.

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Exercise 4.1.1:Pool Connections

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Instructions

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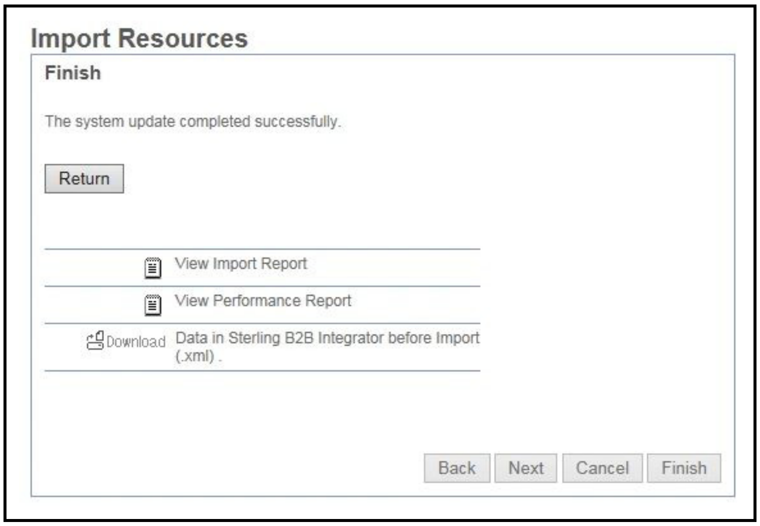
Step 9: Verify confirmation details and click **Finish**. Import Resource page is displayed as shown in the screen capture.



Step 10: Click **Return**.

Exercise 4.1.1: Pool Connections

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Instructions

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Verify Code List Information

Step 1: From the **Administration Menu**, click **Trading Partner** > **Code Lists**.

Step 2: Select **Search by Code List Name** and enter **SWIFT_Currencies**, click **GO!**

Step 3: Select **Source Manager**.

Step 4: Select **edit**, and view the code list of 190 currencies with Sender Code and Receiver code details.

Adding a New Pool Connection

Step 1: From the windows image, click **FileZilla** tool.

Step 2: Click the arrow next to **QuickConnect**.

Step 3: Select **sftp://root@192.168.40.100** to connect to the server system.

Step 4: From Remote site panel access **/opt/IBM/SterlingIntegrator/install/properties**

Step 5: Upload the **jdbc_customer.properties.in** file from c:\6F89G\Labfiles folder into **/opt/IBM/SterlingIntegrator/install/properties.**

Important

To simplify the exercise steps, in creating a new pool. The connection jdbc_customer.properties.in is edited and stored in the C:\6F89G\Labfiles folder. The edited steps are provided in the bullet point after this point.
You need to follow Step 6 to continue with the exercise.

Adding a New Pool Connection

The jdbc_customer.properties.in file is edited with following steps.

Note

Do not perform this step, It is a procedure only.

- Open jdbc_customer.properties.in file in Notepad++ tool.
- You must **copy a block of parameters**, locate the pool settings for. db2Pool.
- Highlight the entire block of setting and copy.
- Paste the db2Pool settings into the **jdbc_customer.properties.in** file in editor tool.

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Exercise 4.1.1: Pool Connections

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Instructions

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Adding a New Pool Connection.

- Select Ctrl + H to replace db2Pool with jdbcService.db2PoolAT. After edit the file must match the screen capture.

```
db2PoolAT.driver=COM.ibm.db2.jdbc.net.DB2Driver
db2PoolAT.driver=com.ibm.db2.jcc.DB2Driver
db2PoolAT.url=jdbc:db2://&DB2_HOST;:&DB2_PORT;/&DB2_DATA;
db2PoolAT.catalog=&DB2_DATA;
db2PoolAT.dbname=&DB2_DATA;
db2PoolAT.varDataClassName=com.sterlingcommerce.woodstock.util.frame.jdbc.DB2VarData
db2PoolAT.user=&DB2_USER;
db2PoolAT.password=&DB2_PASS;
db2PoolAT.maxconn=20;
db2PoolAT.schema=&DB_SCHEMA_OWNER;
db2PoolAT.schema=&DB2_USER;
db2PoolAT.storedProcClassName=com.sterlingcommerce.woodstock.util.frame.jdbc.SybaseStore
db2PoolAT.testOnReserve=true
db2PoolAT.testOnReserveQuery=SELECT PRODUCT_LABEL from DB2INST1.SI_VERSION where PRODUCT
db2PoolAT.testOnReserveInterval=60000
db2PoolAT.maxRetries=100
db2PoolAT.blobPageSize=1024000
db2PoolAT.compressBlob=true
useNewStateAndStatusLogic=true
#db2PoolAT.cacheps=true
#db2PoolAT.cacheps=true
db2PoolAT.dbvendor=db2
db2PoolAT.buffersize=500
db2PoolAT.maxsize=&MAX_TRANS_POOL;
db2PoolAT.initsize=&MIN_TRANS_POOL;
db2PoolAT.factory=com.sterlingcommerce.woodstock.util.frame.jdbc.ConnectionFactory
db2PoolAT.behaviour=2
db2PoolAT.lifespan=0
db2PoolAT.idletimeout=86400000
```

- In the customer_overrides.properties file, edit **db2PoolAT.testOnReserveQuery** parameter adds the schema to the table used in the query. The modified query is as follows.

  `SELECT PRODUCT_LABEL from DB2INST1.SI_VERSION where PRODUCT_LABEL =?`
- Remove the comment for the **db2PoolAT.maxconn** parameter.
- Save the file. The procedure is completed.

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Exercise 4.1.1: Pool Connections

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Instructions

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Adding a New Pool Connection

Step 6: From the PuTTy window, change directory to **/opt/IBM/SterlingIntegrator/install/bin** directory to shut down the system by hardstop command.

```
./hardstop.sh
```

Step 7: Click **Enter**.

Step 8: Enter **./setupfiles.sh** and click **Enter**.

Step 9: Enter **./run.sh** to start the server and click **Enter**.

Step 10: Enter **password** for Passphrase.

Step 11: Click **Enter**.

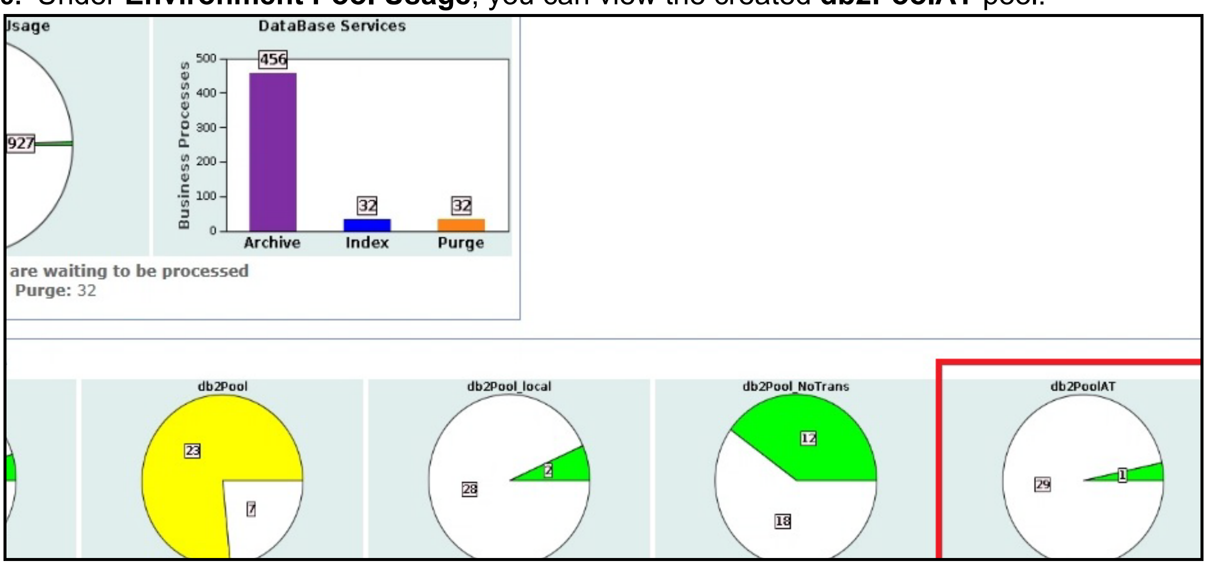
Step 1: From the web browser, access **http://192.168.40.100:9000/dashboard** URL, by specifying **admin** for UserID and **password** for Password.

Step 2: Click **Sign In**.

Step 3: In the **Administration Menu**, select **Operations** > **System** > **Troubleshooter**.

Step 4: Under System Status page, click **Database Usage** link. Click **OK** to proceed.

Step 5: Under **Environment Pool Usage**, you can view the created **db2PoolAT** pool.



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Exercise 4.1.1: Pool Connections

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Instructions

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Creating the Lightweight JDBC Adapter Service Configuration

Step 6: In the **Administration Menu**, select **Deployment** > **Service** > **Configuration**.

Step 7: In the Create section, next to New Services, click **Go!**.

Step 8: In the Service Configuration: Select Service Type page, type **Lightweight JDBC Adapter** and click **Next**.

Step 9: In the Services Configuration: Name page, complete the fields that are listed in the following table by using the parameters that are given and click **Next**.

Field	Value
Name	AT_pool
Description	db2AT pool
Select a group	None

Step 10: In the Service Configuration: AT_pool: Properties page, select **This Lightweight JDBC Adapter will not start a new business process** and click **Next** two times.

Step 11: In the Services Configuration: AT_pool: Properties: Parameters page, complete the fields that are listed in the following table for the parameters given:

Field	Value
Pool Name	db2PoolAT
XML Results Root Tag	ATDocument
XMLResults Row Tag	ATQuery
Query Type	Select
SQL Statement	Select

Step 12: Click **Next**.

Step 13: Verify that the service is enabled for the business processes and click **Finish**.

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Exercise 4.1.1: Pool Connections

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Instructions

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Creating a Business Process to Extract Data Records

Step 1: In the **Administration Menu**, select **Business Processes** > **Manager**.

Step 2: In the Run Graphical Process Modeler section, click **Go!**.

Note

Click Allow and continue to allow blocked note in popup screen.

Step 3: Click **Run**.

Step 4: Enter **admin** for User ID and **password** for Password.

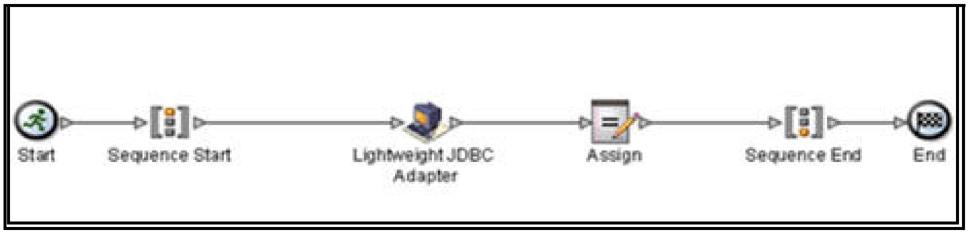
Step 5: Select **View** > **Refresh Services**.

Step 6: Select **File** > **New**.

Step 7: Move the following stencils to your workspace:

Start (1)
End (1)
Sequence Start (1)
Sequence End (1)
Lightweight JDBC Adapter (1)
Assign (1)

Step 8: Arrange and connect the stencils as shown in the following illustration:



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Exercise 4.1.1: Pool Connections

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Instructions

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Creating a Business Process to Extract Data Records

Step 9: Open the Properties Editor for the Lightweight JDBC Adapter and select **AT_pool** for the configuration. Complete the fields in the following table by using the parameters that are given:

Note

To override service configuration, select Options > Preferences > Service Editor tab. Check Override Service Configuration Values check box.

Field	Value
param1	SWIFT_Currencies
paramtype1	String
row_name	ATQuery
sql	select * from CODELIST_XREF_ITEM where list_name = ?
InitialWorkFlowID	[Not Applicable]
pool	db2PoolAT
StartNewWorkFlow	This Lightweight JDBC Adapter will not start a new business process
result_name	ATDocument

Note

The Lightweight JDBC Adapter extracts data records from the SWIFT_Currencies code list and create an XML formatted primary document.

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Exercise 4.1.1: Pool Connections

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Instructions

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Creating a Business Process to Extract Data Records

Step 10: Click the Assign stencil. In the Property Editor – Assign pane, complete the fields that are listed in the following table for the parameters given:

Field	Value
From	DocToDOM
To	Temproot

Note

This assign service moves the Primary Document into Process Data.

Step 11: Save the business process with the name **AT_db2AT**. Save the business process in **C:\6F89G\Labfiles** folder.

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Exercise 4.1.1: Pool Connections

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Instructions

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Check In and Test the Business Process

Step 1: In the **Administration Menu** and chose **Business Process** > **Manager**.

Step 2: In the **Create** section, next to Process Definition, click **Go!**

Step 3: In the Editor: Process Name dialog box, complete the fields in the following table for the parameters given:

Field	Value
Name	AT_db2ATPool Process
Check in Business Process created by the Graphical Modeling Tool	Select

Step 4: Click **Next**. The Check-in-page appears.

Step 5: Browse and open **AT_db2AT.bp**.

Step 6: In the Description text box, type **Using the db2AT pool** and click **Next**.

Step 7: Use the **default setting for the remaining parameters**. Verify that the Business Process is enabled when you click **Finish**.

Testing the Business Process

Step 1: From the **Administration Menu,** select **Business Process** > **Manager**.

Step 2: From the Business Process manager screen, search for the **AT_db2ATpool** business process.

Step 3: Click **execution manager** when you get the results screen.

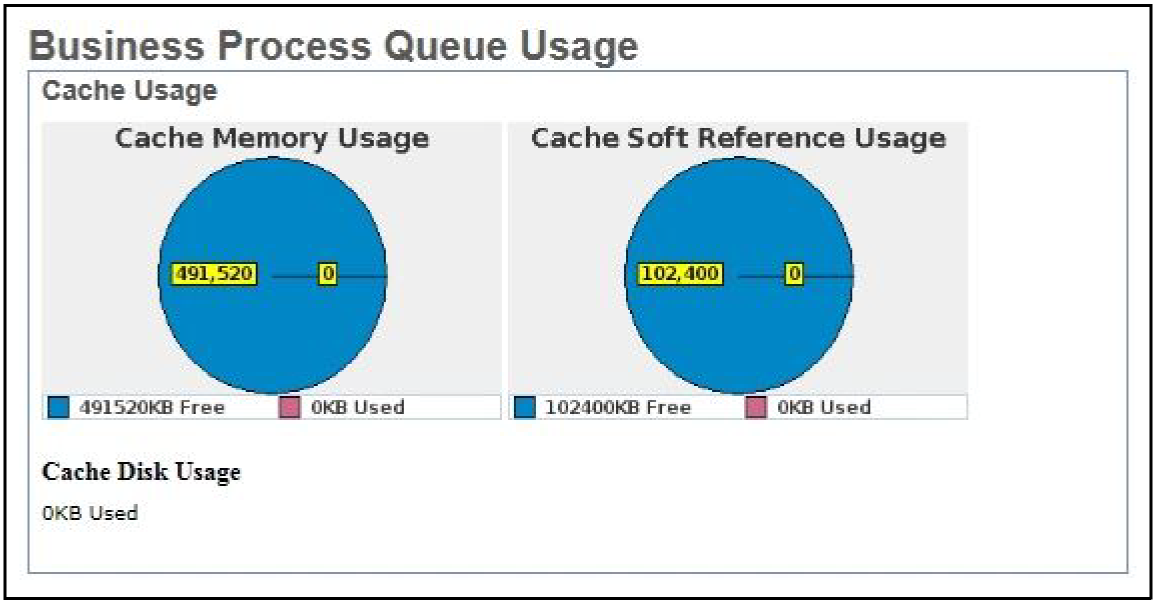
Step 4: Click **execute** in the execution manager screen.

Step 5: Click **Go!**. Observe the separate Execute Business Process window, while the business process runs.

Business Process Queue Usage

View Business Process Queue Usage

The Business Process Queue Usage page allows to diagnose problems with business process queues. To view the business process usage statistics, click **Business Process Queue Statistics** in System Troubleshooting -page. The top of the Business Process Queue Usage page shows the Cache Usage as shown in the following screen capture:



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Business Process Queue Usage

(Continued)

View Business Process Queue Usage

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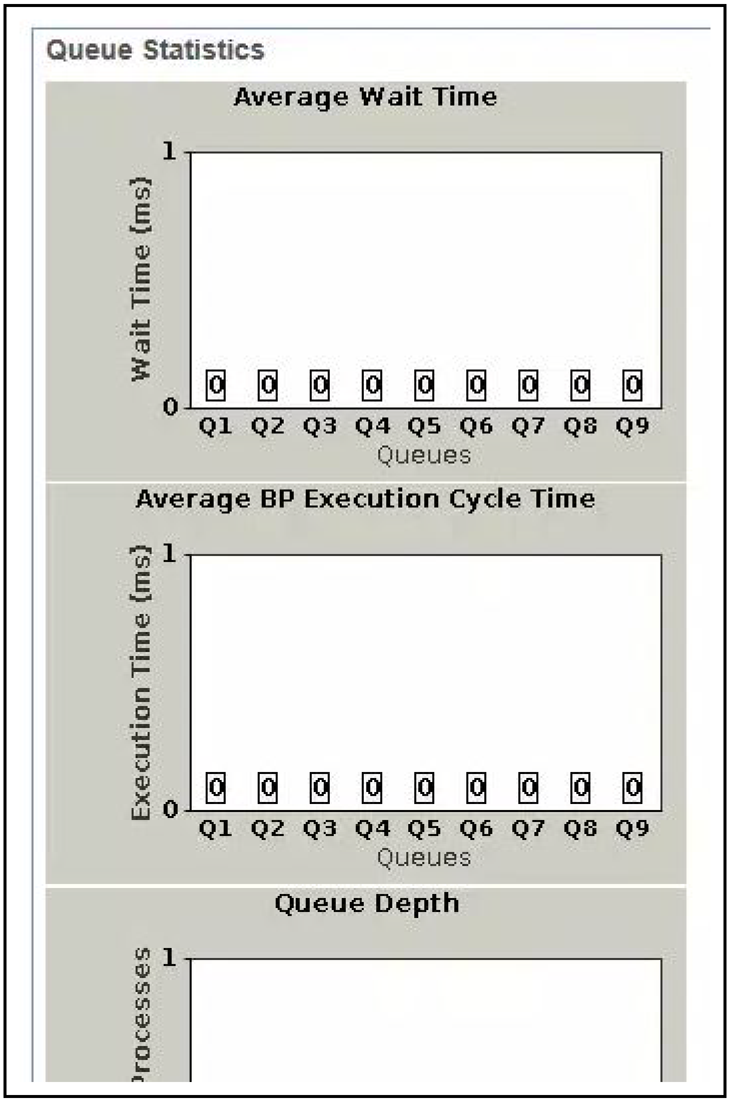
The Queue Statistics is as shown in the following figure:



The Business Process Queue Usage page includes information about the:

- Cache Usage
  - The amount of memory available for cache and the amount consumed.
  - The amount of disk space available for cache and the amount consumed.

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Business Process Queue Usage

(Continued)

View Business Process Queue Usage

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- Queue Statistics
  - The Average Wait Time for business processes per queue.
  - The average business process execution cycle time, which includes the execution times of several steps. It captures the average time that business processes are active on threads before rescheduled. The number of business processes in each queue.
  - Number of business processes within the data size ranges that are processed.
  - The number of business processes that ran without being cached and the number that are currently in cache. BP Queue Usage - Queue Statistics
  - The number of business processes in queues.

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Business Process Queue Usage

(Continued)

View Business Process Queue Usage

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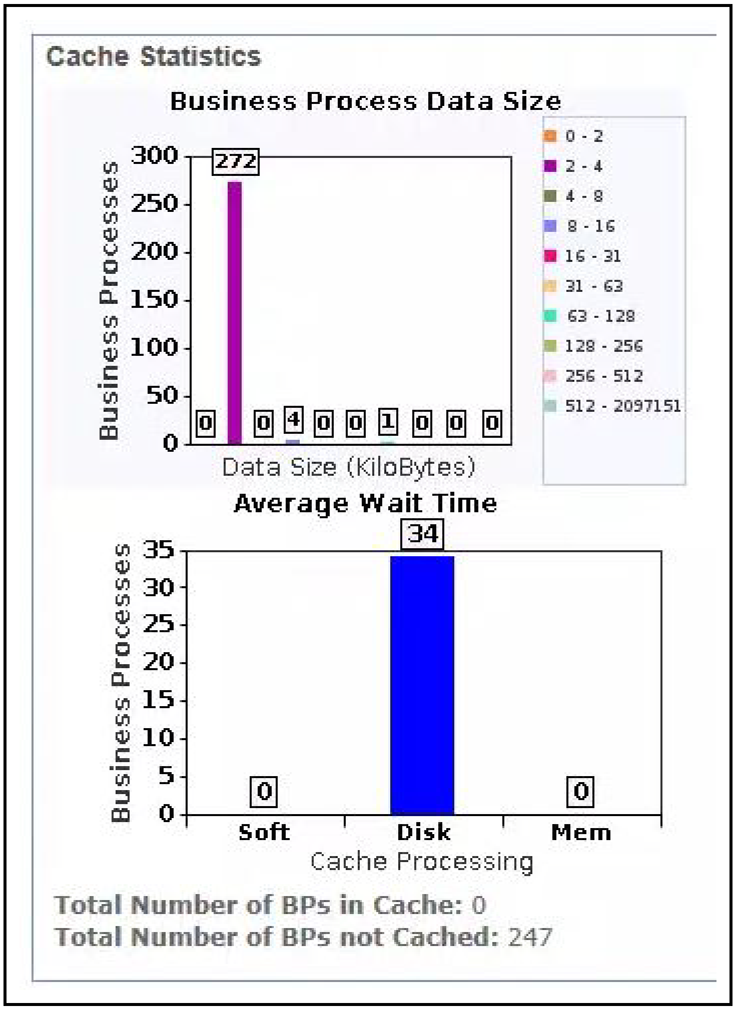
- Cache Statistics

  The Sterling B2B Integrator scheduler access three main context cache locations to speed up processing. Statistics for these three cache locations are reported on the Business Process Usage page under Cache Statistics, as shown in the following screen capture:



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Business Process Queue Usage

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View Business Process Queue Usage

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This Business Process Queue Usage page allows to determine the:

- Number of business processes within the data size ranges that are processed.
- Number of business processes that ran without being cached and the number that are currently in cache.
- Cache location so that you can determine the number of business processes that were found in the Soft Reference Cache, Memory Cache, and Disk Cache.
  - Soft Reference Cache: A soft reference is a kind of Java reference that tells the JVM to “keep this object around you can.” The JVM reclaims these objects if it is running out of memory.
  - Memory Cache: Fixed size memory cache for small objects
  - Disk Cache: On-disk cache with asynchronous deletion

    Processing occurs fastest if the context data retrieved can be found in Soft Reference Cache, followed by Memory Cache and is slowest if it is necessary to retrieve it from Disk Cache.

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Business Process Usage

View Business Process Usage

To see business processes in various states, click BP Usage to access the following Business Process Usage table.The page provides the information about the following business process states.

- ASYNC_QUEUED
- Active
- Halting
- Waiting
- Waiting_On_IO
- Interrupted_Man
- Interrupted_Auto

Business Process Usage

State	Process Count
Halted_Softstop	0
ASYNC_QUEUED	0
Active	1
Halted	10
Halting	0
Waiting	0
Waiting_On_IO	0
Interrupted_Man	18
Interrupted_Auto	0

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Business Process Usage

(Continued)

View Business Process Usage

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You can click a Process Count number to see all business processes in that state and control business process execution.

Choices for active status are restart, stop, and expedite. Choices for interrupted status are resume, restart, and terminate.

Business Process Issues

You can find that the number of business processes in a halting, halted, waiting, or interrupted state increases. These states indicate that either performance is inadequate and corrective action is required, or you have business processes that have errors and need manual attention. Symptoms of an increasing number of business processes in a halting, halted, waiting, or interrupted state includes:

- Slows system performance
- Database getting full or having performance issues
- Business processes complete with errors, which places them in a halted state
- Business Process Usage report shows an increasing number of business processes in a halted, halting, waiting, or interrupted state

Resolve Business Process Issues

Halted, Halting, Waiting, or Interrupted business processes

In Sterling B2B Integrator, business processes that fail receive a state of halted. The failed business processes are not placed in a completed state, because if Sterling B2B Integrator gave a failed business process a state of completed, the failed business processes must be archived or purged without corrective action taken. Giving a failed business process a state of halted enables to take manual corrective action without the business process archived or purged.

- Halted or interrupted business processes: A business process in a Halting, Halted, Interrupted_Man, or Interrupted_Auto state requires immediate attention because the business process is stopped processing. In Sterling B2B Integrator, business processes remain in a halted or interrupted state until some action is taken on the business process.

  When you notice a halted or interrupted business process, you have two options:

  - Terminate the business process.
  - Restart the business process.
- Waiting business processes: A business process in a waiting state might not require immediate attention; the business process can write on a resource or a document from another business process.

Note

If you notice numerous business processes in a waiting state, these states indicate a performance issue that requires attention.

When you notice a business process in a waiting state, you have three options:

- Allow the business process to remain in the waiting state if it is waiting on resources or a service or activity that is disabled, but is enabled
- Terminate the business process
- Restart the business process

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Resolve Business Process Issues

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Potential Causes of Halted, Waiting, or Interrupted Business Processes

Determining the cause of an increasing number of business processes in a halting, halted, waiting, or interrupted state might require you to investigate many areas of the system and how you are implementing Sterling B2B Integrator.

Causes of an increasing number of business processes in a halting, halted, waiting, or interrupted state includes:

- System, business process, or activity schedules are disabled.

Example

if a business process requires an output from, or access to, a different service or business process that is scheduled to work, but the schedule is not turned on, the places the business process in a halted or waiting state.

- System errors.

Example

Java, JVM, out of memory errors, or operating system errors might cause a business process to halt or be interrupted. Check your business process logs for causes of the halted or interrupted business processes. If the logs show JVM errors, Java errors, or operating system errors, review your operating system documentation for resolutions.

- Sterling B2B Integrator stops running.

Example

Your site experiences a power outage and you must restart Sterling B2B Integrator after power is restored. The business processes at the time of the power outage might be in halted, interrupted, or waiting states after the recovery operations run, depending on the activities completed at the time of the outage.

Determining the Cause of Halted, Waiting, or Interrupted Business Process

To determine the cause of an increasing number of business processes in a halting, halted, waiting, or interrupted state, review:

- The Business Process Usage report on the **Operations** > **System** > **Troubleshooting** page. This report shows the number of business processes in the different states. Click the number next to the state to view detailed information about the process.

(Continued on next page)

Resolve Business Process Issues

(Continued)

Determining the Cause of Halted, Waiting, or Interrupted Business Process

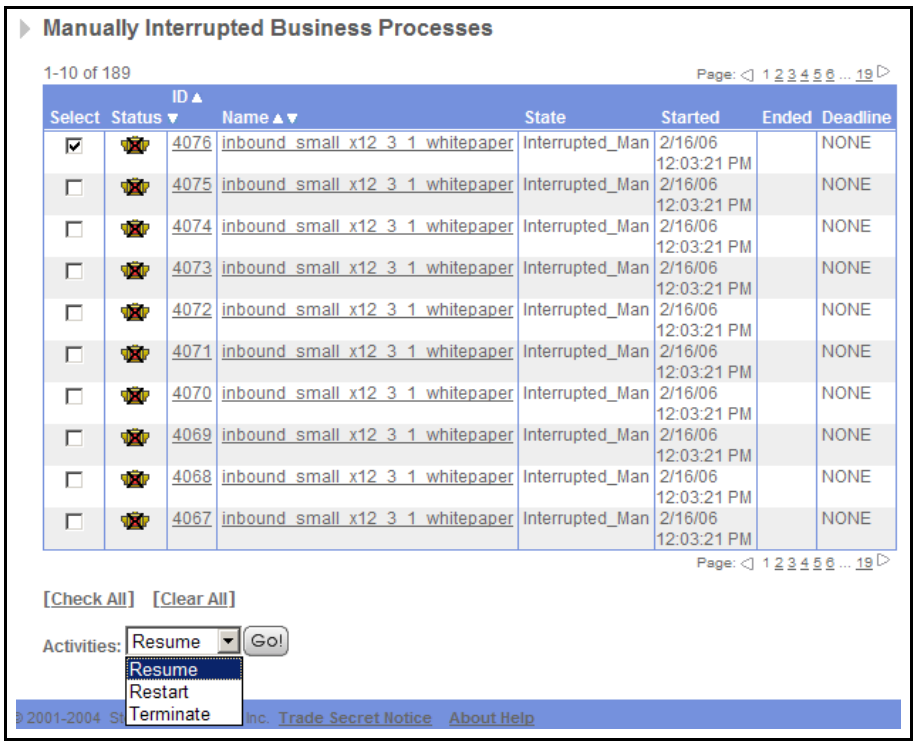
....(Continued)

- Appropriate system and business process schedules to verify that they are turned on.

Example

if you notice that many business processes are halting and each of these business processes depend on the schedule of another business process or service, this activity indicates that the scheduled business process or service might not be turned on.

- The Performance Statistics report for information that is related to the business process execution times. Increasing execution times for key business processes or activities indicate that a business process is not efficiently designed, or that a resource leak might occur.



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Classpath

Introduction

You can view the system classpath for debugging purposes and to verify whether third-party libraries are available in the classpath.

The classpath is an environment variable that tells the Java compiler javac.exe where to look for class files to import or java.exe where to find class files to interpret. Click classpath on the System Troubleshooting page in the System Status area. The screen capture lists few of the classpath of Sterling B2B Integrator application.

Note

Sometimes library conflicts where two different versions of the same package are installed and the wrong one is being used. Another problem is when an application is looking for a library and it is not in the path.


Classpath

System Class Path

/opt/IBM/SterlingIntegrator/install/jar/bootstrapper.jar -- file

Dynamic Class Loader

```
/opt/ibm/db2/V10.5/java/db2jcc.jar -- file
/opt/IBM/SterlingIntegrator/install/dbjar/jdbc/DB2/db2jcc.jar -- file
/opt/IBM/SterlingIntegrator/install/properties/ -- directory
/opt/IBM/SterlingIntegrator/install/noapp/deploy/filegateway/webapp/WEB-INF/lib/isomorphic_core_rpc.jar
/opt/IBM/SterlingIntegrator/install/jdk/jre/lib/xml.jar -- file
/opt/IBM/SterlingIntegrator/install/jar/commons_collections/3_2_2/commons-collections-3.2.2.jar -- file
/opt/IBM/SterlingIntegrator/install/jar/jakarta_jstl/1_1_2/standard.jar -- file
/opt/IBM/SterlingIntegrator/install/jar/jakarta_jstl/1_1_2/jstl.jar -- file
/opt/IBM/SterlingIntegrator/install/jar/jasper/5_5_15/jasper-runtime-5.5.15.jar -- file
/opt/IBM/SterlingIntegrator/install/jar/jasper/5_5_15/jasper-compiler-jdt-5.5.15.jar -- file
/opt/IBM/SterlingIntegrator/install/jar/jasper/5_5_15/jasper-compiler-5.5.15.jar -- file
/opt/IBM/SterlingIntegrator/install/jar/platform_baseutils.jar -- file
/opt/IBM/SterlingIntegrator/install/jar/install_foundation.jar -- file
/opt/IBM/SterlingIntegrator/install/jdk/lib/tools.jar -- file
/opt/IBM/SterlingIntegrator/install/jar/log4j/1_2_15/log4j-1.2.15.jar -- file
/opt/IBM/SterlingIntegrator/install/jar/ant/1_7_1/ant.jar -- file
/opt/IBM/SterlingIntegrator/install/jar/ant/1_7_1/ant-junit.jar -- file
/opt/IBM/SterlingIntegrator/install/jar/ant/1_7_1/ant-nodeps.jar -- file
/opt/IBM/SterlingIntegrator/install/jar/ant/1_7_1/ant-launcher.jar -- file
/opt/IBM/SterlingIntegrator/install/jar/ant/1_7_1/ant-trax.jar -- file
/opt/IBM/SterlingIntegrator/install/jar/javamail/1_4/mail.jar -- file
/opt/IBM/SterlingIntegrator/install/jar/javamail/1_4/dsn.jar -- file
/opt/IBM/SterlingIntegrator/install/jar/javamail/1_4/mailapi.jar -- file
/opt/IBM/SterlingIntegrator/install/jar/javamail/1_4/pop3.jar -- file
```

JNDI Tree

Introduction

You can view the system JNDI Tree for debugging purposes. Verify whether the expected resources.

Example

adapters or pool names, are in the JNDI tree. Click JNDI Tree on the System Troubleshooting page in the System Status area. The system JNDI Tree displays showing the JNDI name and class name pairs.

View a Node JNDI Tree in a Clustered Environment

You can view a specific node JNDI Tree for debugging purposes and to verify whether the expected resources, for example, adapters or pool names, are in the JNDI tree. This option is available only in a clustered environment.

In the System Status area, click node#, where # is the number of the node you want to view information about. The node JNDI Tree displays showing the JNDI name, class name pairs, and the node name as shown in the screen capture.


JNDI Tree

Name	Class Name
B2B_FTP_CLIENT_ADAPTER_B2B_FTP_CLIENT_ADAPTER_node1	com.sterlingcommerce.woodstock.services.ftpclient.FtpClientSer
B2B_SIB_ADAPTER_B2B_SIB_ADAPTER_node1	com.sterlingcommerce.woodstock.services.sib.SibServerImpl
B2B_SMTP_CLIENT_ADAPTER_B2B_SMTP_CLIENT_ADAPTER_node1	com.sterlingcommerce.woodstock.services.b2bsmtp.B2BMailSM
BackupServiceType_node1	com.sterlingcommerce.woodstock.services.backup.BackupAdapt
BSF_node1	com.sterlingcommerce.woodstock.services.bsf.BSFServerImpl
CDServerBPFaultLogger_BPFaultLog_node1	com.sterlingcommerce.woodstock.services.bpfaultlog.BPFaultLo
E5Logger_BPFaultLog_node1	com.sterlingcommerce.woodstock.services.bpfaultlog.BPFaultLo
EBICSClientHTTPClientAdapter_HTTPClientAdapter_node1	com.sterlingcommerce.woodstock.services.httpclient.HttpClientA
EBICSHttpServerAdapter_HttpServerAdapter_node1	com.sterlingcommerce.woodstock.services.pshttp.PSHttpAdapte
ebXMLHttpServerAdapter_HttpServerAdapter_node1	com.sterlingcommerce.woodstock.services.pshttp.PSHttpAdapte
eInvoiceLogger_BPFaultLog_node1	com.sterlingcommerce.woodstock.services.bpfaultlog.BPFaultLo
FileSystem_node1	com.sterlingcommerce.woodstock.services.filesystem.FileSyster
FTPClientAdapter_FTPClientAdapter_node1	com.sterlingcommerce.woodstock.services.psftpclient.FtpClientA
FTPSend_FTP_SEND_ADAPTER_node1	com.sterlingcommerce.woodstock.services.ftpclient.FtpClientSer
FTPSendToGISSupport_FTP_SEND_ADAPTER_node1	com.sterlingcommerce.woodstock.services.ftpclient.FtpClientSer

Clean-Up Processes Monitor

Clean-Up Processes Monitor

You can view how long the processes are available, since the completion of different cleanup processes, including archiving, purging, and indexing. This status helps you see on one screen whether these processes are running and completing.

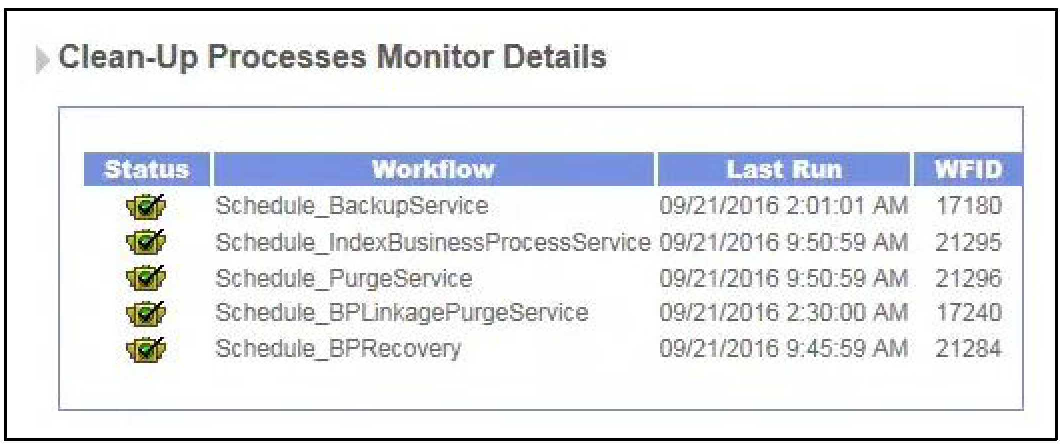
To view when a cleanup process is completed:

1. From the **Administration Menu**, select **Operations** > **System** > **Troubleshooter**.
2. In the **System Status** area, click **Clean-Up Processes Monitor**.

   The Clean-Up Processes Monitor Details window displays, showing the status, workflow (or cleanup process) name, the date, and time when the workflow was last run, and the workflow ID as shown in the screen capture.

   The Status column has the following values:

   - Red – More than four times the scheduled interval elapsed without a successful execution.
   - Yellow – More than three times the scheduled interval elapsed without a successful execution.
   - Green – The last scheduled instance that is completed successfully.
   - Gray – The process is never completed any scheduled instance or it is never scheduled.


Clean-Up Processes Monitor Details

Status	Workflow	Last Run	WFID
	Schedule_BackupService	09/21/2016 2:01:01 AM	17180
	Schedule_IndexBusinessProcessService	09/21/2016 9:50:59 AM	21295
	Schedule_PurgeService	09/21/2016 9:50:59 AM	21296
	Schedule_BPLinkagePurgeService	09/21/2016 2:30:00 AM	17240
	Schedule_BPRecovery	09/21/2016 9:45:59 AM	21284

Cache Usage

Viewing Cache Usage

Sterling B2B Integrator uses caches to hold information that is frequently requested by the system. To change cache settings, see the Performance Tuning Utility. To view the cache usage statistics, in the Application Status area, click Cache Usage.

The Cache Usage report shows the following information for each cache type:

Cache Usage

1-25 of 32 Page: 1 2

Name	Count	Requests	Hits
YCPTemplateManager	51	1437	1386
dbclasses/YFS_PROCESS_TYPE/list	1	5	4
dbclasses/YFS_RESOURCE/list	1	6	5
dbclasses/YFS_USER/select	3	109	106
simplCache	501	7557	4646
dbclasses/YFS_SHIP_NODE/select	1	5	4
dbclasses/YFS_ORGANIZATION/select	1	5	4
dbclasses/YFS_INSTALL_DEFAULTS/list	2	2	0
dbclasses/YFS_USER_EXIT_IMPL/list	2	2	0
dbclasses/YFS_BASE_COMMON_CODE/select	1	5	4
dbclasses/YFS_USER_GROUP/select	2	30	28
dbclasses/YFS_RESOURCE/select	258	1292	1034
dbclasses/YFS_BASE_CONFIG_GRP_DEFN/select	1	5	4
PermissionCache	1	26	25
dbclasses/YFS_APPLICATION_MENU/select	1	5	4
dbclasses/YFS_LOCALIZED_STRINGS/list	1	1	0
workflowdef	35	4146	4110
dbclasses/YFS_ENTERPRISE/select	1	5	4
dbclasses/YFS_MENU/list	164	820	656
dbclasses/YFS_BASE_CONFIG_TABLES/select	1	179	178
dbclasses/YFS_USER_GROUP_LIST/list	1	15	14
XPathCacheLoader	41	9847	9801
UserCache	2	1794	1790
dbclasses/PLT_PROPERTY/list	2	6	4
dbclasses/YFS_TRANSACTION/select	1	4	3

(Continued on next page)

Cache Usage

(Continued)

Viewing Cache Usage

....(Continued)

The Cache Usage page includes information about the:

- Cache name – Name of the cache
- Count – Number of objects in the cache
- Requests – Number of times an object was requested from the cache, regardless of success.
- Hits – Number of times an object was requested from the cache and found.

Note

The cache settings in tuning.properties for translation maps, Envelopes, and Other EDI affects the size of the corresponding cache areas here.

Resolutions for Inefficient Use of Cache

The following resolutions are for the inefficient use of cache, depending on the identified cause:

- Improperly Tuned Cache Performance Properties
- If you review the Cache Usage report and notice the number of requests that are increasing and the number of hits that are decreasing for the same cache, increase the cache values with the Performance Tuning Utility or manually increase the values in the install_dir/properties/cacheManager.properties file.
- Trade offs in increasing cache sizes include:
- Increasing the cache sizes too much can make the system no more effective than reading from the disk. Depending on your system and your configuration, you might need to adjust the cache settings to attain peak performance without losing the benefit of cache over disk.
- Increasing the cache sizes for items that are not used frequently can degrade performance because more resources are allocated to the caches, but are not being used.

(Continued on next page)

Cache Usage

(Continued)

Resolutions for Inefficient Use of Cache

....(Continued)

- High Cache Allocation for Less Frequently Used Large Objects

  If you review the Cache Usage report and notice the number of requests to a cache are low, you can be able to reduce the size of the cache values with the Performance Tuning Utility, or manually decrease the values in the install_dir/properties/cacheManager.properties.

  The low request number indicates that the objects in the cache are not used frequently with your business processes. Review the value for the cache property and reduce the cache size if the cache size is large.

  The trade-off in decreasing cache sizes is that decreasing the cache sizes too much can cause a reduced number of hits to the caches, which cause a call to the database for the data and increases the processing times.

  Depending on your system and your configuration, you might need to adjust the cache settings to attain peak performance without losing the benefit of cache over disk.

(Continued on next page)

Cache Usage

(Continued)

Resolutions for Inefficient Use of Cache

....(Continued)

- Low Cache Allocation for More Frequently Used Small Objects

  If you review the Cache Usage report and notice the number of requests to a cache are high and the number of hits to the cache are low, you might be able to increase the size of the cache values with the Performance Tuning Utility, or manually increase the values in the install_dir/properties/cacheManager.properties file.

  The high request number indicates that the objects in the cache are used frequently with your business processes.

  Review the value for the cache property and increase the cache size if the cache size is small. This activity is especially important for the smaller objects that are used frequently and are static in value. Cache retrieval is faster than disk or database retrieval.

  The trade-off in increasing cache sizes for smaller objects is that increasing the cache sizes too much can cause a reduced number of hits to the caches for other caches and objects.

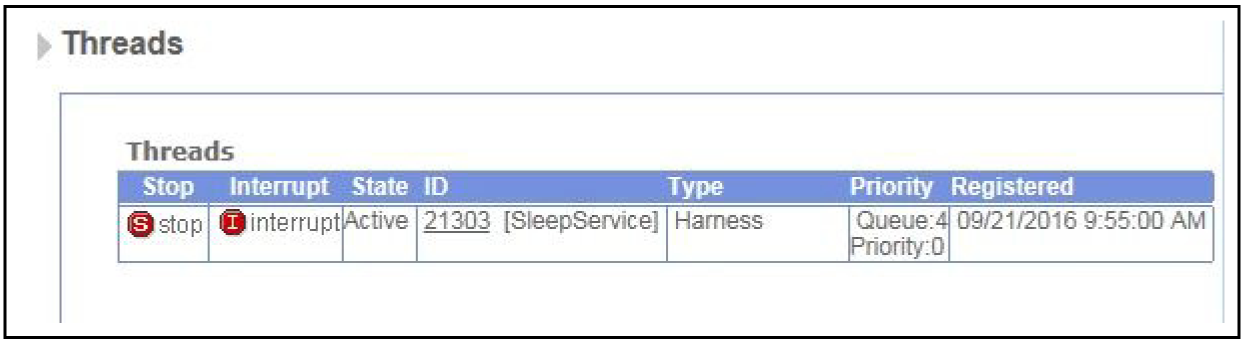
  The call to the database for the data and increases the processing times. Depending on your system and your configuration, you might need to adjust the cache settings to attain peak performance without losing the benefit of cache over disk.

View Threads

Introduction

To view threads in Sterling B2B Integrator, in the Application Status area, click Threads.

The Threads report shows any threads on the system and their status.



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Controllers

Refreshing a Controller

You can refresh controllers in your environment with the System Troubleshooting page. To refresh a controller, in the Application Status area, next to the controller you want to refresh, click the refresh icon in the Refresh column.

The service controller can be refreshed when the configuration changes for a stateful adapter. The refresh must test all setup and initialization necessary for the adapter to function correctly.

If an adapter maintains a connection or connections to an end system, and the connectivity configuration changes, the refresh returns a false value to the service controller.

In this case, the service controller shuts down the adapter and then restarts it. This action prevents workflows from trying to start the adapter while connectivity changes are occurring. This method returns a Boolean value.

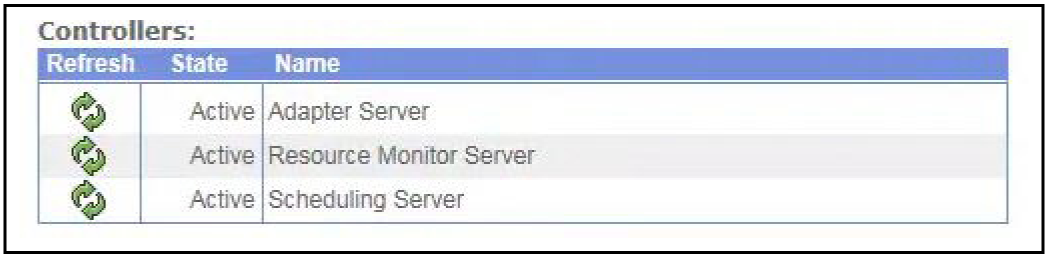
A true return indicates that the refresh that is completed successfully. A false return indicates that the adapter did not refresh, in which case the service controller shuts down the adapter and then restarts it.

Adapter Controllers

Introduction

You can view the adapters by its state that are currently active or stopped, to verify accuracy or to plan changes as needed.

Adapters can be enabled or disabled in the System Troubleshooting page. Clicking the adapter name displays its service settings, characteristics, and status details.



Perimeter Servers

Creating Perimeter Servers

You can follow the steps to create a new perimeter server:

1. Select Operations > Perimeter Servers from the Administration Menu. The Perimeter Servers page is displayed.
2. Click add to create a new perimeter server. The Perimeter Server Configuration page is displayed. The following table describes the fields in perimeter server configuration page.

Field	Description
Name	Name that you provided of the perimeter server to connect to.
Near End Configuration	
Interface Or IP	DNS name or IP address of the computer that you typed when you installed the perimeter server. Type * to allow Sterling Integrator to establish this value. This interface is used for the near end of the persistent connection to the perimeter server. Specify it only if your system has multiple interfaces and not all are able to connect to your DMZ. **Note**: Do not use in iSeries (OS/400) environments.

Note

The Near End Configuration fields are useful in environments involve firewalls with rules that are designed to allow specific IP addresses, ports, or both to create outbound connections. However, it is not allowed in iSeries (OS/400) environments.

(Continued on next page)

Perimeter Servers

(Continued)

Creating Perimeter Servers

....(Continued)

Field	Description
Local Port	Port number that you chose when you installed the perimeter server. Type 0 (zero) to allow Sterling Integrator to establish this value. This port is used for the near end of the persistent connection to the perimeter server. Specify a port other than 0 (zero) only if your firewall controls access to the DMZ based on the originating port. Specifying 0 (zero) allows Sterling B2B Integrator to choose any available port. **Note**: Do not use in iSeries (OS/400) environments.
Perimeter Server (far-end) is in less secure network zone	Check to enable the connection from Sterling Integrator to the perimeter server. To connect in the opposite direction, clear the checkbox.
Perimeter Server Host	DNS name or TCP/IP address of the computer that the remote perimeter server is installed on. If you specified an internal interface during your perimeter server installation, use that address here.
Perimeter Server Port	Port number that the remote perimeter server monitors for connections. is the port number that you specified during remote perimeter server installation.
Cluster Node	Node that is to be used with this perimeter server, if you are running in a clustered environment. If you are running in a clustered environment, you must select the node from the list.

(Continued on next page)

Perimeter Servers

(Continued)

Creating Perimeter Servers

....(Continued)

3. Clicking Next the **High/Low Watermark** page is displayed. The following table describes the fields in the High/Low Watermark page.

Field	Description
Inbound Connection	High: Highest inbound connection buffer size is the high watermark. When a trading partner sends data faster than Sterling Integrator can process. The excess data accumulates inside perimeter services in the inbound connection buffer. The buffer size reaches the High Inbound Connection value.The perimeter services stop receiving data for that connection until enough of excess data is processed by inbound connection buffer size drops to the Low Inbound Connection value. For example, if you set the High Inbound Connection value to 500 KB and the Low Inbound Connection value to 250 KB, perimeter services stop receiving data when the inbound connection buffer size reaches 500 KB and resumes receiving data when the inbound connection buffer size drops to 250 KB.
	Low: Lowest inbound connection buffer size is the low watermark.

Perimeter Servers

(Continued)

Creating Perimeter Servers

....(Continued)

4. Clicking Next the **High/Low Watermark** page is displayed. The following table describes the fields in the High/Low Watermark page.

Field	Description
Outbound Connection	High: Highest outbound connection buffer size is the high watermark. When Sterling Integrator sends data to a trading partner faster than the trading partner can receive it, the excess data accumulates inside perimeter services in the outbound connection buffer. The buffer size reaches the High Outbound Connection value, perimeter services stop sending data through that connection until enough of the excess data is sent that the outbound connection buffer size drops to the Low Outbound Connection value. For example, if you set the High Outbound Connection value to 500 KB and the Low Outbound Connection value to 250 KB, perimeter services stop sending data when the outbound connection buffer size reaches 500 KB and resumes sending data when the outbound connection buffer size drops to 250 KB.
	Low: Lowest outbound connection buffer size is the low watermark.

(Continued on next page)

Perimeter Servers

(Continued)

Creating Perimeter Servers

....(Continued)

5. Click Next and verify the perimeter server settings in the Confirm page.
6. Click the Finish to confirm the settings. The perimeter server is added to Sterling Integrator. You can now monitor the perimeter server by using the Troubleshooter page.

Note

The property settings for perimeter servers are stored in the Properties directory of the Sterling Integrator installation.

The following properties files are affected by the perimeter servers settings:

- perimeter.properties
- log.properties

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Perimeter Servers

(Continued)

Perimeter Server Status

To monitor the existing perimeter servers:

- Select the Operations > System > Troubleshooter from the Administration Menu. The System Troubleshooting page is displayed.
- Select the Perimeter Server Status link. The Perimeter Server window opens with the details of available perimeter servers.

The Perimeter Servers window displays the following information about the perimeter servers. If you are working in a clustered environment, the information that displays is mandated by the node you select from the Select Node list.

- Name of the cluster node that the perimeter server is associated - Cluster Node name (in a clustered environment only)
- State – Enabled or Disabled - Whether the perimeter server is on or off.
- Name of the perimeter server - Name of the perimeter server
- Last Activity - Date and time of the last activity the perimeter server used

View and Enable/Disable Perimeter Server

To enable a perimeter server, in the Perimeter Servers window, next to the perimeter server that you want to enable, in the On/Off column, select the checkbox. To disable the Perimeter server, clear the check box.

You can view the perimeter server settings to verify accuracy or to plan changes as needed. To view perimeter server settings, in the Name column, click the name of the perimeter server that you want to view.

Note

The local or perimeter client/server is displayed in the Sterling B2B Integrator installation for this class and cannot be enabled, disabled, or viewed by using Troubleshooter. A true perimeter server or remote perimeter server must be installed to view its information.

System Troubleshooting - Databases Services

Introduction

If the Sterling B2B Integrator database is installed on this server, the Database Services area displays the following information.

- Host - Name of the host on which a database is installed.
- Location - Location or path of the database installation.
- State - State of the database, either active or inactive.

Important

The name for this area displays showing the name of the active database.

System Troubleshooting - RMIAPrime Services

Introduction

If RMIAPrime services are in use, the RMIAPrime Services area shows the following information:

- Host - Name of the host
- Location - Location or path to the RMIAPrime Service
- State - State of the RMIAPrime Service, either active or inactive

Important

These menus are not included in all Systems.

(Continued on next page)

Monitor Cluster Node Status

Node Status information

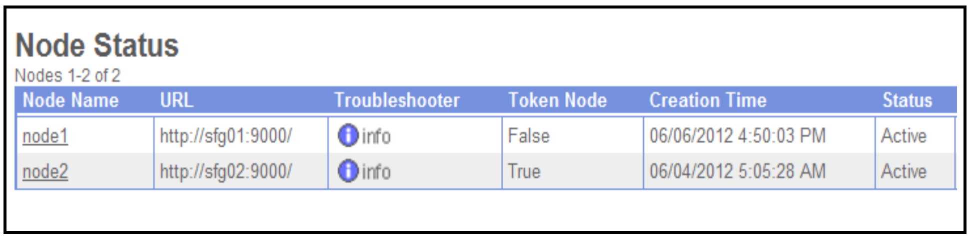
If you are working in a clustered environment, information is available about all the nodes in the cluster. The lab environment is of non-clustered environment.

To view node status information

From the Administration Menu, select Operations > System > Cluster > Node Status. The Node Status page shows the following information:

Note

Only in Cluster environment you will see this report.



Node Status

Nodes 1-2 of 2

Node Name	URL	Troubleshooter	Token Node	Creation Time	Status
node1	http://sfg01:9000/	info	False	06/06/2012 4:50:03 PM	Active
node2	http://sfg02:9000/	info	True	06/04/2012 5:05:28 AM	Active

The following table describes the Node Status page:

Column	Description
Name	Shows the name of the node. Click the name to view more details about the node.
URL	Shows the Uniform Resource Locator for the node.
Troubleshooter	Provides a link to the System Troubleshooter.

Monitor Cluster Node Status

(Continued)

Node Status information

Column	Description
Token Node	States whether it is the token node. Valid values include: True – is the token node. False – is not the token node.
Creation Time	Shows the date and time that the node was created.
Status	Shows the status of the node. Valid values include: Active – Node is working and available for processing Starting Ops – Node is starting up, but not available for processing Node went down – Node is not working and not available for processing

If a node went down, in the install_dir/bin directory of the node, run the hardstop.sh (UNIX) or hardstop.cmd (Windows) script and then run the run.sh (UNIX) or run.cmd (Windows) script. It stops all processing and restarts the node.

(Continued on next page)

Monitor Cluster Node Status

(Continued)

View Node Status Detail

To view more details about a node, click the Name of the node on the Node Status page. A page Appears that contain the information about the node, as shown in the following figure.

node1

Node Name	node1
URL	http://sfg01:9000/
Token	False
Creation Time	06-06-2012 16:50:03
Status	Active
BasePort	9000
Location	/opt/apps/sfg1_9000/install/noapp
Role	Default
Operation Controller Host	sfg01
Operation Controller Port	9027
SI_Install_Base	5201
PLATFORM	6000
licensed_product	6000
platform_afc_security	601204
platform_afc	601204
platform_ifc	1030300

(Continued on next page)

Monitor Cluster Node Status

(Continued)

View Node Status Detail

....(Continued)

The following table describes some of the key detailed information for each node:

Information	Description
Location	Shows the directory path where the node is installed.
Role	States the role of the node.
Operation Controller Host	Shows the name of the server that is acting as the Operation Controller Host.
Operation Controller Port	Shows the port number for the Operation Controller.
Sterling B2B Integrator version	Shows which version of Sterling B2B Integrator is used for the node.
JVM Version	Shows the version number for the Java Virtual Machine.
JVM Vendor	Shows the vendor that provided the Java Virtual Machine.

Lesson review

What you have been able to do

After completing this lesson, you should have been able to:

- View Database Usage, Services, and Pools.
- View business process Queue, Usage statistics, and State information.
- Resume, restart, or terminate a business process.
- View system classpath and JNDI tree information.
- View the cleanup process.
- View environment statistics, including cache and memory used.
- View Remote Perimeter Server information if present.
- View Troubleshooter information about Sterling B2B Integrator cluster nodes.
- Monitor node information and threads with Cluster Monitor.